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Inventor(s) Colgrove; James R. et al.

Apparatuses, methods, and systems for vibratory screening

Abstract

Vibratory screening machines that include stacked screening deck assemblies are provided. In some embodiments, at least one of the vibratory screening machines can include an outer frame, an inner frame connected to the outer frame, and a vibratory motor assembly secured to the inner frame for vibrating the inner frame. A plurality of screen deck assemblies can be attached to the inner frame in a stacked arrangement, each configured to receive replaceable screen assemblies. The screen assemblies can be secured to respective ones of the plurality of the screen deck assemblies by tensioning the screen assemblies in a direction that a material to be screened flows across the screen assemblies. An undersized material discharge assembly can be configured to receive materials that pass through the screen assemblies, and an oversized material discharge assembly can be configured to receive materials that pass over the screen assemblies.

Inventors: **Colgrove; James R. (East Aurora, NY), Peresan; Michael L. (Strykersville, NY)**

Applicant: **Derrick Corporation** (Buffalo, NY)

Family ID: **61902937**

Assignee: **Derrick Corporation (Buffalo, NY)**

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Primary Examiner: Matthews; Terrell H

Attorney, Agent or Firm: FisherBroyles, LLP

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 17/387,644, filed Jul. 28, 2021, which is a continuation of U.S. patent application Ser. No. 17/002,219, filed Aug. 25, 2020, now U.S. Pat. No. 11,779,959, which is a continuation of U.S. patent application Ser. No. 16/513,963, filed Jul. 17, 2019, now U.S. Pat. No. 10,773,278, which is a divisional of U.S. patent application Ser. No. 15/785,141, filed Oct. 16, 2017, now U.S. Pat. No. 10,399,124, which claims the benefit of U.S. Provisional Patent Application No. 62/408,514, filed Oct. 14, 2016, and U.S. Provisional Patent Application No. 62/488,293, filed Apr. 21, 2017, the entire contents of each of which are incorporated herein by reference.

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a perspective side view of a vibratory screening machine, according to one or more embodiments of the present disclosure;
(2) FIG. **2** is a perspective top view of the vibratory screening machine shown in FIG. **1**;
(3) FIG. **3** is a front view of the vibratory screening machine shown in FIGS. **1** and **2**;
(4) FIG. **4** is a rear view of the vibratory screening machine shown in FIGS. **1**, **2**, and **3**;
(5) FIG. **5** is an isometric view of a screening deck having screen assemblies mounted thereon,

according to one or more embodiments of the present disclosure;
(6) FIG. **6** is an enlarged partial isometric view of the screening deck shown in FIG. **5**, without screen assemblies mounted thereon, incorporated into the vibratory screening machine shown in FIGS. **1**, **2**, **3**, and **4**;
(7) FIG. **7** is an enlarged side view of a wash tray, which may be incorporated into the screening deck shown in FIGS. **5** and **6**, according to one or more embodiments of the present disclosure;
(8) FIG. **8** is an isometric view of a tensioning device with a ratchet mechanism, according to one or more embodiments of the present disclosure;
(9) FIG. **9**A is a side view of the screening deck shown in FIGS. **5**, **6**, and **7** with the ratchet mechanism shown in FIG. **8**;
(10) FIG. **9**B is an enlarged view of the ratchet mechanism shown in FIG. **9**A;
(11) FIG. **10** is an enlarged partial isometric view of a feed assembly and the screening deck shown in FIGS. **5**, **6**, and **7** secured to the vibratory screening machine shown in FIGS. **1**, **2**, **3** and **4**;
(12) FIG. **11**A is an isometric bottom view of an undersized material discharge assembly, according to one or more embodiments of the present disclosure;
(13) FIG. **11**B is an isometric top view of the undersized material discharge assembly shown in FIG. **11**A;
(14) FIG. **12**A is an isometric bottom view of an oversized material discharge chute, according to one or more embodiments of the present disclosure;
(15) FIG. **12**B is an isometric top view of the oversized material discharge chute shown in FIG. **12**A;
(16) FIG. **13**A is an isometric top view of an oversized material discharge trough, according to one or more embodiments of the present disclosure;
(17) FIG. **13**B is an isometric bottom view of the oversized material discharge trough shown in FIG. **13**A, according to one or more embodiments of the present disclosure;
(18) FIG. **14** is a cross-sectional side view of a screening deck having material flowing across the screening deck and featuring an impact area of a screen assembly incorporated into a screening deck assembly, according to one or more embodiments of the present disclosure;
(19) FIG. **15** a side view of a tray showing material to be filtered falling on an impact area of a filter member, according to one or more embodiments of the present disclosure.
(20) FIG. **16**A is a front-side perspective view of a screen assembly, according to one or more embodiments of the present disclosure.
(21) FIG. **16**B is a side view of a screen filter for use in an embodiment of the present disclosure.

Description

DETAILED DESCRIPTION
(1) The present disclosure relates generally to methods and apparatuses for screening materials, in particular, for separating materials of varying sizes. Embodiments of the present disclosure include screening systems, vibratory screening machines, and apparatuses for vibratory screening machines and screen assemblies for separating materials of varying sizes.
(2) Vibratory screening systems are disclosed in U.S. Pat. Nos. 6,431,366 B2 and 6,820,748 B2, which are incorporated herein by reference thereto. Advantages of the present invention over previous systems include a larger screening capacity for separation of materials without an associated increase in machine size. Embodiments of the present invention include improved features such as: screening deck assemblies having first and second screens; tensioning devices that tension each screen in a front to back direction (i.e., in the direction of flow of the material that is being screened); wash trays positioned in between the first and second screens; feed chutes configured to connect directly to an over-mounted feed system, e.g., the feed systems described in

U.S. Patent App. No. 2014/0263103 A1, which is incorporated herein by reference hereto; centralized discharge assemblies which collect undersized and oversized materials; and replaceable screen assemblies configured for front to back tensioning and impact areas for flow of material onto the screen assemblies. These features, among others described herein, provide for a compact design that allows for a direct overhead feed system, increased screening capacity, and reduced footprint. Additionally, the multiple screen assemblies that are tensioned front to back with wash trays in between and impact areas on the screen assemblies themselves provide for improved flow characteristics and efficiencies. The improved tensioning structures provide for quick and easy replacement of screen assemblies. The improved discharge assemblies are configured for optimal or nearly optimal flow characteristics as well as for providing the greatly reduced footprint. These improvements and advantages, and others, are provided by at least some embodiments in accordance with aspects of this disclosure.

(3) Example embodiments of the present disclosure employ vibratory screening machines to separate materials of varying sizes. In some embodiments, a vibratory screening machine includes a framing assembly, a plurality of screening deck assemblies mounted to the framing assembly, an undersized material discharge assembly and an oversized material discharge assembly. The framing assembly includes an inner frame mounted to an outer frame. A plurality of screening deck assemblies are mounted to the inner frame and arranged in a stacked and staggered relationship. Each screening deck assembly includes a first screening deck and a second screening deck, a wash tray extending between first and second screening decks, and a tensioning assembly. At least one vibrating motor may be attached to the inner frame and/or at least one screening deck assembly. An undersized material discharge assembly and an oversized material discharge assembly, each of which may include at least one vibratory motor, are in communication with each screening deck assembly, and are configured to receive undersized and oversized screened material, respectively, from the screening deck assemblies.

(4) In one embodiment of the present disclosure, a vibratory screening machine includes an outer frame, an inner frame connected to the outer frame, a vibratory motor assembly secured to the inner frame such that it vibrates the inner frame. A plurality of screen deck assemblies is attached to the inner frame in a stacked arrangement, each configured to receive replaceable screen assemblies. The screen assemblies are secured to the screen deck assemblies by tensioning the screen assemblies in a direction that a material to be screened flows across the screen assemblies. An undersized material discharge assembly is configured to receive materials that pass through the screen assemblies, and an oversized material discharge assembly is configured to receive materials that pass over a top surface of the screen assemblies. The undersized material discharge assembly includes an undersized chute in communication with each of the screen deck assemblies and the oversized material discharge assembly includes an oversized chute assembly in communication with each of the screen deck assemblies.

(5) The oversized chute assembly may include a first oversized chute assembly and a second oversized chute assembly. The undersized chute, the first oversized chute assembly, and the second oversized chute assembly may be located beneath the plurality of screen deck assemblies, and the undersized chute may be located between the first and second oversized chute assemblies. At least one of the plurality of screen deck assemblies may be replaceable. Each screen deck assembly may include a first screen assembly and a second screen assembly. A wash tray may be located between the first screen assembly and the second screen assembly. A trough may be located between the first screen assembly and the second screen assembly. The trough may include an Ogee-weir structure.

(6) The vibratory screening machine may include a screen tensioning system that includes tensioning rods that extend substantially orthogonal to the direction of flow of the material being screened. The tensioning rods may be configured to mate with a portion of the screen assembly and tension the screen assembly when rotated. The screen tensioning system may include a ratcheting

assembly configured to rotate the tensioning rod such that it moves between a first open screen assembly receiving position to a second closed and secured screen assembly tensioned position.
(7) The vibratory screening machine may include a vibratory motor, wherein the vibratory·motor is attached to the oversized chute assembly. The vibratory screening machine may include multiple feed assembly units, each feed assembly unit located substantially directly below individual discharges of a flow divider. The vibratory screening machine may include at least eight screen deck assemblies.
(8) The oversized chute assembly may include a bifurcated trough that is configured to receive materials that do not pass through the screen assemblies and are conveyed over a discharge end of the screen deck assemblies. A first section of the bifurcated trough may feed the first oversized chute assembly, and a second section of the bifurcated trough may feed the second oversized chute assembly.
(9) In one embodiment of the present disclosure, a screen deck assembly includes a first screen deck configured to receive a first screen assembly, a second screen deck configured to receive a second screen assembly located downstream from the first screen deck assembly; and a trough located between the first and second screen deck assemblies, wherein the first screen deck assembly is configured to receive a material to be screened and the trough is configured to pool the material to be screened before it reaches the second screen deck assembly.
(10) The trough may include at least one of an Ogee-weir and a wash tray. The screen deck assembly may include a first and a second screen tensioning system, each having tensioning rods that extend substantially orthogonal to the direction of flow of the material to be screened. The first tensioning rod may be configured to mate with a first portion of the first screen assembly when rotated and the second tensioning rod may be configured to mate with a second portion of the second screen assembly when rotated.
(11) The first screen tensioning system may include a first ratcheting assembly configured to rotate the first tensioning rod such that the first tensioning rod moves between a first open screen assembly receiving position to a second closed and secured screen assembly tensioned position. The second screen tensioning system may include a second ratcheting assembly configured to rotate the second tensioning rod such that the second tensioning rod moves between a first open screen assembly receiving position to a second closed and secured screen assembly tensioned position.
(12) In one embodiment of the present disclosure, a method of screening a material includes feeding the material on a vibratory screening machine having a plurality of screen deck assemblies that are configured in a stacked arrangement, each of the screen deck assemblies configured to receive replaceable screen assemblies, the screen assemblies secured to the screen deck assemblies by tensioning the screen assemblies in the direction the material flows across the screen assemblies; and screening the materials such that a undersized material that passes through the screen assemblies flows into an undersized material discharge assembly, and an oversized material flows over an end of the screen deck assembly into an oversized material discharge assembly. The undersized material discharge assembly includes an undersized chute in communication with each of the screen deck assemblies and the oversized material discharge assembly includes an oversized chute assembly in communication with each of the screen deck assemblies.
(13) The oversized chute assembly may include a first and second oversized chute assembly. The undersized chute and first and second oversized chute assemblies may be located beneath the plurality of screen deck assemblies, and the undersized chute may be located between the first and second oversized chute assemblies.
(14) At least one of the plurality of screen deck assemblies may be replaceable. Each screen deck assembly may include a first and a second screen assembly. A. trough may be located between the first and second screen assemblies. The trough may include an Ogee-weir structure.
(15) A screen tensioning system may be included having tensioning rods that extend substantially

orthogonal to the direction of flow of the material being screened. The tensioning rods may be configured to mate with a portion of the screen assembly and tension the screen assembly when rotated.
(16) FIGS. **1** to **4** illustrate a vibratory screening machine **100**. Vibratory screening machine **100** includes a framing assembly having an outer frame **110**, and an inner frame **120**, a feed assembly **130**, a plurality of screening deck assemblies **400**, a top vibratory assembly **150**, an undersized collecting assembly **160** and an oversized collecting assembly **170**.
(17) FIG. **1** illustrates a side perspective view of vibratory screening machine **100**. FIG. **2** illustrates a top perspective view of vibratory screening machine **100**, shown from the opposite side of vibratory screening machine **100** as is illustrated in FIG. **1**. As is shown in FIG. **2**, the opposite side of vibratory screening machine **100** includes mirror image components of outer frame **110** as is shown in FIG. **1**. The mirror-image outer frame components are denoted by the addition of a prime (′) at the end of the corresponding component reference number.
(18) As is shown in FIGS. **1** and **2**, outer frame **110** includes a longitudinal set of base supports **111** and **111′**, a latitudinal set of base supports **112** and **112′**, and two sets of upstanding channels, **113** and **113′** and **114** and **114′**. Upstanding channels **113** and **113′** and **114** and **114′** each have first ends **113**A and **113′**A and **114**A and **114′**A, mid-portions **113**B and **113′**B and **114**B and **114′**B, and second ends **113**C and **113′**C and **114**C and **114′**C, respectively. Each of first ends **113**A and **113′**A and **114**A and **114′**A are elevated relative to second ends **113**C and **113′**C and **114**C and **114′**C, with mid-portions **113**B and **113′**B and **114**B and **114′**B extending the length between the first and second ends, respectively. Outer frame **110** further includes upper angled channels **115** and **115′** and lower angled channels **116** and **116′**. Upper angled channels **115** and **115′** and lower angled channels **116** and **116′** each have first ends **115**A and **116**A, mid-portions **115**B and **116**B, and second ends **115**C and **116**C, respectively. First ends **115**A and **116**A are elevated relative to second ends **115**C and **116**C, and mid-portions **115**B and **116**B extend the length between first ends **115**A and **116**A and second ends **115**C and **116**C, respectively. Outer frame **110** also includes three sets of declining channels: **117** and **117′**, **118** and **118′**, and **119** and **119′**. Each declining channel has a first end, **117**A, **118**A, and **119**A which is elevated relative to its respective second end, **117**B, **118**B, **119**B.
(19) Referring to FIGS. **1** and **2**, the opposite ends of longitudinal base supports **111** and **111′** attach to the opposite ends of latitudinal base supports **112** and **112′** such that the four base supports create a rectangular shape. Second ends **113**C and **113′**C and **114**C and **114′**C of each respective upstanding channel attach to the four corners where base channels **111** and **111′** meet base channels **112** and **112′**. Mid-portion **113**B and **113′**B of upstanding channel **113** attaches to first end **119**A of declining channel **119**. Second end **119**B of declining channel **119** rests above longitudinal base support **111**. First end **113**A of upstanding channel **113** attaches to mid-portion **115**B of upper angled channel **115** and first end **118**A of declining channel **118**. First end **115**A of upper angled channel **115** attaches to first end **117**A of declining channel **117**. Second end **117**B of declining channels **117** attaches to mid-portion **116**B of lower angled channel **116** towards first end **116**A. Second end **118**B of declining channel **118** attaches to mid-portion **116**B of lower angled channel **116** toward second end **116**C. Second end **116**C of lower angled channel **116** attaches to and terminates at second end **119**B of declining channel **119**.
(20) Referring to FIG. **2**, outer frame **110** further includes a rear channel **109** having opposite ends that attach to one of each of mid-portions **113**B and **113**B′ of upstanding channel **113**. Additional rear channels **108** run parallel to rear channel **109**, each with opposite end attached to lower angled channel **116** and its counterpart lower angled channel **116′** from mid-portion **116**B toward second end **116**C to provide structural support to outer frame **110**.
(21) As is shown in FIG. **2**, inner frame **120** mounts top vibratory assembly **150** and screening deck assemblies **400** via securing mechanisms, such as bolts. Inner frame **120** includes upper angled channels **125** and **125′**, lower angled channels **126** and **126′**, upper declining channels **127** and **127′**,

and lower declining channels **128** and **128′**. Upper and lower angled channels **125** and **126** of inner frame **120** run parallel to upper and lower angled channels **115** and **116** on the medial side of outer frame **110**. Upper and lower declining channels **127** and **128** of inner frame **120** run parallel to declining channels **117** and **118** on the medial side of outer frame **110**. Though not shown in FIGS. **1** and **2**, inner frame **120** may be mounted to outer frame **110** with elastomeric mountings, or other similar mountings, that permit inner frame **120** to maintain vibratory motion while dampening the effects of vibration on the structural integrity of fixed outer frame **110**. In an embodiment, elastomeric mountings are made of a composite material including rubber and have female threads that accept male bolts from the inner frame and outer frame. The elastomeric mountings may be replaceable parts. While outer frame **110** is shown in the specific configuration described, it may have different configurations as long as it provides the structural support necessary for inner frame **120**. In embodiments, vibratory screening machine **100** may have an outer frame that includes feet that are configured to attach to an existing structure.

(22) In some embodiments, top vibratory assembly **150** includes side plates **153** and **153′**, a first vibrating motor **151**A and a second vibrating motor **151**B. Side plates **153** and **153′** have a top angled edge **154**, a bottom edge **155**, and an exterior surface **156**. Bottom edge **155** of side plate **153** is secured to a side channel **430** of screening deck assembly **400** via securing mechanisms, such as bolts. Exterior surface **156** includes ribs **157** that provide structural support to top vibratory assembly **150**. The opposing sides of vibrating motor **151**A and second vibrating motor **151**B are mounted to top angled edges **154** of side plates **153** and **153′**. First and second vibrating motors **151**A and **151**B are configured such that they may vibrate all screening deck assemblies **400** mounted to inner frame **120**. While shown with a particular configuration in FIGS. **1** and **2**, it is noted that top vibratory assembly **150** may have other arrangements that retain the functionality described herein.

(23) As is shown in FIG. **2**, vibratory screening machine **100** includes a feed assembly **130**. Feed assembly **130** includes support frame **134**, a plurality of vertical supports **136**, feed inlet ducts **131**, mounting arms **132**, and feed outlet ducts **133**. Mounting arms **132** are secured to support frame **134** and **134′** with securing mechanisms, such as bolts. Support frame **134** and **134′** is located above and parallel to declining channels **117** and **117′** of outer frame **110**. Vertical supports **136** secure support frame **134** and **134′** to declining channels **117** and **117′** of outer frame **110** such that feed assembly **130** is fixed relative to vibrating inner frame **120**. Inlet ducts **131** are configured to receive a flow of slurry from a flow divider device, such as shown in U.S. Patent Application No. 2014/0263103 A1, which is incorporated herein by reference in its entirety, or other material flow assemblies, and feed it to outlet ducts **133**. Outlet ducts **133** are positioned above elevated sides of screening deck assemblies **400** such that each outlet duct **133** is configured to discharge a flow of materials **500** to each screening deck assembly **400**. Earlier systems have hoses located a story above vibratory machines, whereas in assemblies of this disclosure, configurations of inlets on the vibratory machine provide for substantially distributed drops in flow and greatly reduce the height of the machine. This is an important space saving feature of at least some embodiments of the present disclosure.

(24) FIG. **3** illustrates a front view of the vibratory screening machine **100**. FIG. **4** illustrates a rear view of the vibratory screening machine **100**. As is shown in FIGS. **3** and **4**, the vibratory screening machine **100** includes an undersized material collection assembly **160** and an oversized material collection assembly **170**. Referring to FIG. **3**, undersized material collection assembly **160** includes a plurality of collecting pans **161** secured to the underside of each screening deck assembly **400**, a plurality of ducts **162** in communication with collecting pans **161**, and an undersized collecting chute **166**. Oversized material collection assembly **170** includes a plurality of oversized collecting chutes **171** mounted to lower end plate **428** of each screening deck assembly **400**, and two oversized collecting troughs **176** and **176′** in communication with oversized collecting chutes **171**. As is shown in FIG. **4**, oversized collecting troughs **176** and **176′** include vibratory motors **179** and

179′. As is shown in FIGS. **3** and **4**, undersized collecting chute **166** extends between oversized collecting chute **171** and oversized collecting troughs **176** and **176**′ beneath screening deck assemblies **400** of vibratory screening machine **100**. Though shown in a specific configuration, oversized collecting troughs **176** and **176**′ and vibratory motors **179** and **179**′ may have different arrangements so long as they aid in conveying oversized material **500** discharged from screening deck assemblies across oversized collecting troughs **176** and **176**′.

(25) FIGS. **5** to **10** illustrate various views of a screening deck **400**. FIG. **5** illustrates an enlarged isometric perspective view of screen assembly **400**. Screening deck assembly **400** includes a first screening deck **410**, a second screening deck **420**, side channels **430** and **430**′, a wash tray **440**, and a tensioning device **450**. As is shown in FIG. **5**, first screening deck **410** and second screening deck **420** are covered by a first screen assembly **409** and a second screen assembly **419**, respectively. First screen assembly **409** and second screen assembly **419** are replaceable screen assemblies which are attached to first and second screening decks **410** and **420**. When in operation, material to be screened **500** by vibratory screening machine **100** is discharged from feed outlet ducts **133** of feed assembly **130** to the elevated side of first screen assembly **409**, along feed end **409**A of first screen assembly **409**, and is vibrated across first screen assembly **409** of first screening deck **410**, over discharge end **409**B of first screen assembly **409**, and into wash tray **440**. Vibration carries material **500** over wash tray **440**, where material passes over feed end **419**A of second screen assembly **419**. As is described herein, material **500** hits second screen assembly **419** in screen impact area **448**, then vibrates across second screen assembly **419** of second screening deck **420**, and over discharge end **419**B of second screen assembly **419** along lower end plate **428**. First screen assembly **409** and second screen assembly **419** are configured such that undersized materials fall through first screen assembly **409** and second screen **419** into undersized material collecting pans **161**, and are funneled into undersized collecting chute **166** via ducts **162**. Oversized materials do not pass through screens **409** and **419** and are vibrated off lower end plate **428** and funneled through oversized collecting chutes **171** and **171**′ to oversized collecting troughs **176** and **176**′. Direction of the flow of material is represented with large arrows. While illustrated in this particular configuration in the figures, oversized collecting chutes **171** and **171**′ and oversized collecting troughs **176** and **176**′ may have different arrangements so long as they receive oversized materials discharged from each screening deck assembly and provide functionality as described herein. The flow of material through split outside oversized collecting chutes **171**, **171**′ and a central undistributed undersized collecting chute **166** provides for efficient flows in reduced space. The configuration of the chutes **166**, **171**, **171**′ reduces the footprint of the machine **100** while providing for direct and efficient flow.

(26) First screening deck **410** includes an upper end plate **416** and a lower end plate **418**. Second screening deck **420** includes an upper end plate **426** and a lower end plate **428**. Opposite sides of first screening deck **410** and second screening deck **420** are secured to the medial sides of side channels **430** and **430**′ with securing mechanisms such as, e.g., bolts or welding. The lateral sides of side channels **430** and **430**′ include a plurality of angled plates **432**. Angled plates **432** include holes through which securing mechanisms, such as bolts, may extend to secure side channels **430** and **430**′ to upper declining channel **127** and **127**′ and lower declining channel **128** and **128**′ of inner frame **120**. While illustrated in this particular arrangement, side channels **430** and **430**′ and angled plates **432** may have different configurations so long as they permit screening deck assembly **400** to vibrate such that materials **500** of varying sizes are separated as desired.

(27) FIG. **6** illustrates a partial side perspective view of screening decks **410** and **420**, wash tray **440**, side channel **430**, and a portion of tensioning device **450**. As is shown in FIG. **6**, a flexible material **405** covers outlet duct **133** of feed assembly **130**. Flexible material **405** is configured to control the flow of materials from outlet duct **133** to screening deck assembly **400** so that the flow of material is uniformly distributed across screening deck assembly **400**, thereby maximizing efficiency of vibratory screening machine **100**. As is shown in FIG. **6**, first screening deck **410** and

second screening deck **420** do not include screens **409** and **419**, but it will be appreciated that first and second screening decks **410** and **420** are covered by screens **409** and **419** when vibratory screening machine **100** is employed to separate materials of varying sizes, and can be changed out, as described herein, when worn or damaged. Referring to FIG. **6**, first screening deck **410** includes a rib **412**, stringers **414**, an upper end plate **416** and a lower end plate **418**. Second screening deck **420** includes a rib **422**, stringers **424**, an upper end plate **426** and a lower end plate **428**. Opposite ends of ribs **412** and **422** extend from side channel **430** and **430**′ at each of the midpoints between upper end plate **416** and lower end plate **418** of first screening deck **410**, and upper end plate **426** and lower end plate **428** of second screening deck **420**, respectively. A plurality of stringers **414** and **424** extend from upper end plates **416** and **426** to lower endplates **418** and **428**, respectively. A midpoint **415** of each stringer **414** and a midpoint **425** of each stringer **424** traverses the top surface of ribs **412** and **422**. Midpoints **415** and **425** are elevated with respect to opposite ends of stringers **414** and **424** such that stringers **414** and **424** create a “crown” or curvature across first and second screening decks **410** and **420**. Though first screening deck **410** and second screening deck **420** are shown with a single rib **412** and **422** respectively, it will be appreciated that first screening deck **410** and second screening deck **420** may include other configurations. First screening deck **410** and second screening deck **420** may include, respectively, a first plurality of ribs and a second plurality of ribs, so long as the additional ribs provide the functionality as described herein. In some embodiments at least one (or, in some embodiments, each one) of the first plurality of ribs and the second plurality of ribs can be assembled similarly to rib **412** or rib **422**.

(28) Distinct from screening assemblies of other systems, such as those disclosed in U.S. Pat. No. 6,431,366, stringers **414** and **424** may be replaceable units, and may be bolted to ribs **412** and **422** rather than welded to ribs **412** and **422**. This configuration eliminates closely spaced weld joints between ribs **412** and **422** and stringers **414** and **424** that are commonly found in welded screening decks. This arrangement eliminates the shrink, heat distortion and drop associated with closely spaced weld joints, and enables rapid replacement of worn or damaged stringers **414** and **424** in the field. Replaceable stringers **414** and **424** may include plastic, metal, and/or composite materials and may be constructed by casting and/or injection molding. While not shown in FIG. **6**, screening decks **410** and **420** are configured to support screens **409** and **419**, which extend across the surface of first screening deck **410** and second screening deck **420**, covering ribs **412** and **422** and stringers **414** and **424**, respectively, as is shown in FIG. **5**.

(29) With further reference to FIG. **6**, upper end plate **416** of first screening deck **410** is elevated relative to lower end plate **418**. Similarly, upper end plate **426** of second screening deck **420** is elevated relative to lower end plate **428**. Wash tray **440** extends between lower endplate **418** of first screening deck **410** and upper endplate **426** of second screening deck **420**. First screening deck **410**, wash tray **440**, and second screening deck **420** are configured such that a flow of material from outlet duct **133** and flexible material **405** of feed assembly **130** traverses first screening deck **410** and wash tray **440** before traversing second screening deck **420**. This configuration enables a flow of materials to be effectively separated by increasing the surface area on which the flow of materials is screened into oversized material collecting assembly **170** and undersized material collecting assembly **160** without increasing the footprint of vibratory screening machine **100**.

(30) FIG. **7** illustrates an isometric side view of wash tray **440** interfacing with first screening deck **410** and second screening deck **420**. As is shown in FIG. **7**, wash tray **440** includes an upper side member **442** having a top portion **442**A and a bottom portion **442**B, a lower member **444** having a first end **444**A and a second end **444**B, and a curved side member **446** including a first end **446**A and a second end **446**B. Curved side member **446** includes an S-shape curve referred to as an “Ogee,” discussed in more detail below. Top portion **442**A of upper side member **442** connects to lower end plate **418** of first screening deck **410**. Bottom portion **442**B of upper side member **442** connects to first end **444**A of lower member **444**. Second end **444**B of lower member **444** connects to first end **446**A of curved side member **446**. Second end **446**B of curved side member **446** curves

over upper end plate **426** of second screening deck **420**.

(31) The resulting configuration of wash tray **440** generates a weir **447**, which is a trough or depression that provides a structure for pooling a flow of liquid or slurry material to be screened **500**. Embodiments of a wash tray **440** having an Ogee-weir structure possess functional significance in the field of fluid dynamics. An Ogee-weir structure is generally described as slightly rising up from the base of a weir and reaching a maximum rise **449** at the top of the S-shaped curve of the Ogee structure. Upon or after reaching maximum rise point **449**, fluid falls over the Ogee structure in a parabolic form. The discharge equation for an Ogee-weir is:

(32) $Q = \frac{2}{3}C_d \times L\sqrt{2g(H)^{\frac{3}{2}}}$

(33) As is shown in FIG. **7**, incorporating wash tray **440** with an Ogee-weir curved side member **446** between first screening deck **410** and second screening deck **420** of screening deck assembly **400** may direct the flow of material screened by first screening deck **410** onto a desired impact point or impact area **448** near upper end plate **426** of second screening deck **420**, or another desired location, such that the discharge flow impacts the downstream screen panel at a predetermined wear surface as opposed to non-uniformly impacting downstream screen surfaces such as the screen openings. In this configuration, impact point/area **448** may remain unchanged despite changes in fluid parameters such as, e.g., flowrate and/or viscosity. Incorporation of Ogee-weir shaped curved side member **446** into wash tray **440** improves screening efficiency and consistency and reduces wear on second screening deck **420**. Flows of materials after impact are represented with large arrows in FIG. **7**.

(34) FIGS. **8**, **9**A and **9**B illustrate tensioning device **450**. FIG. **8** illustrates an isometric perspective view of tensioning device **450**. Tensioning device **450** includes a tensioning rod **451**, brackets **454** and **454′**, and ratchet mechanisms **456** and **456′**. FIG. **9**A illustrates a partial side view of two ratchet mechanisms **456** and two brackets **454** mounted to side channel **430** of screening deck assembly **400**. FIG. **9**B illustrates an enlarged view of one of two ratchet mechanisms **456** and brackets **454** shown in FIG. **9**A. As described in more detail below, each screening deck assembly **400** includes two tensioning devices **450**, one configured to enable tensioning of screen assembly **409** of first screening deck **410**, and the other configured to enable tensioning of screen **419** of second screening deck **420**.

(35) Referring to FIG. **8**, tensioning device **450** includes a tensioning rod **451**, brackets **454** and **454′**, and ratchet mechanisms **456** and **456′**. Tensioning rod **451** includes opposing, mirror image ends **452** and **452**,′ a tubular midportion **453**, and a tensioning strip **455**. Opposing ends **452** and **452′** of tensioning rod **451** extend through holes **457** and **457′** in ratchet mechanisms **456** and **456′**, respectively, and are secured to ratchet mechanisms **456** and **456′** by securing mechanisms, such as bolts. Ratchet mechanisms **456** and **456′** are secured to brackets **454** and **454′**, which are in turn secured to side channels **430** and **430′**, respectively, of screening deck assembly **400**, by securing mechanisms, such as bolts, as is shown in FIGS. **9**A and **9**B.

(36) While not shown in FIG. **8**, tubular mid-portion **453** of tensioning rod **451** extends the width of screening deck assembly **400** from side channel **430** to side channel **430′**. Tensioning rods **451** of each tensioning device **450** are located beneath upper end plate **416** of first screening deck **410** and upper end plate **426** of second screening deck **420**. Tubular mid-portion **453** and tensioning strip **455** of tensioning device **450** are configured to receive an end of screen assembly **409** and/or **419**. Opposing end **452**, tubular mid-portion **453**, and tensioning strip **455** of tensioning rod **451** are arranged so that when opposing end **452** and tubular mid-portion **453** rotate in a counter-clockwise direction, tensioning strip **455** rotates in a clockwise direction, thereby pulling screen assembly **409** and/or **419** towards upper end plate **416** of first screening deck **410** and/or upper end plate **426** of second screening deck **420**. While shown in FIG. **8** as having tubular mid-portion **453** and tensioning strip **455**, tensioning device **450** may include other components so long as it is configured receive an end of screen assembly **409** and/or **419** and is connected to ratchet

mechanism **456** so as to permit ratchet mechanism **456** to rotate tensioning rod **451** and pull screen assembly **409** and/or **419** toward upper end plates **416** and/or **426**.

(37) FIG. **9**A illustrates a partial side view of two ratchet mechanisms **456** and two brackets **454** of two tensioning devices **450** mounted to side channel **430** of screening deck assembly **400**. FIG. **9**B illustrates an enlarged view of ratchet mechanism **456** and bracket **454**. Though not shown, tensioning rods **451** extend from each ratchet mechanism **456** on side channel **430** of screening deck assembly **400** to each ratchet mechanism **456**′ on opposing side channel **430**′ beneath upper end plates **416** and **426** of screening deck assembly **400**.

(38) FIG. **10** illustrates an enlarged partial perspective view of ratchet mechanism **456** mounted to side channel **430** below first screening deck **410**. First screening deck **410** is shown interfacing with feed assembly **130** and flexible flow controlling material **405**. As is shown in FIG. **10**, ratchet mechanism **456** includes an upper portion **458** and a lower portion **460**. Upper portion **458** includes a locking bar **459** that interfaces with a multitude of teeth **461** on lower portion **460**. Lower portion **460** includes an actuation point **462** where second end **452** of tensioning rod **451** extends through hole **457** of ratchet mechanism **456**. Referring to FIG. **10**, a wrench **463** is configured to rotate actuation point **462** of ratchet mechanism **456**. In response to application of a counter-clockwise rotational force to wrench **463**, actuation point **462** and tubular mid-portion **453** of tensioning rod **451** are configured to rotate in a counter-clockwise direction, and tensioning strip **455** is configured to rotate in a clockwise direction such that tensioning device **450** pulls an end of screen assembly **409** toward upper end plate **416**. In response to rotation of wrench **463** and actuation point **462** of ratchet mechanism **456**, locking bar **459** of upper portion **458** and teeth **461** of lower portion **460** are configured to lock the tensioning device in place and retain tension. Whereas tensioning devices used in vibratory screening machines disclosed in the prior art apply tension in a side-to-side direction, or towards side channels **430** and **430**′ relative to vibratory screening machine **100**, tensioning device **450** disclosed herein applies tension in a front-to-back direction, or towards upper end plate **416** and lower end plate **418** of first screening deck **410** and/or upper end plate **426** and lower end plate **428** of second screening deck **420** relative to vibratory screening machine **100**. Unlike tensioning devices disclosed in the prior art, the front-to-back direction of tensioning provided by tensioning device **450** corresponds with the direction of the flow of material such as, e.g., slurry, across first and second screening decks as it is separated by vibratory screening machine **100**. Though shown with wrench **463** in FIG. **10**, other tools may be employed to rotate actuation point **462** of ratchet mechanism **456**, so long as it provides functionality as described herein.

(39) FIGS. **11**A and **11**B illustrate an embodiment of undersized material collection assembly **160**. Undersized material collection assembly **160** includes a plurality of collecting pans **161** secured to the underside of each screening deck assembly **400** (see FIGS. **3** and **4**), a plurality of ducts **162** in communication with collecting pans **161**, and an undersized collecting chute **166**. As is shown in FIGS. **11**A and **11**B, undersized collecting chute **166** includes a mounting end **167**, which may be secured to outer frame **110** of vibratory screening machine **100** by securing mechanisms, such as bolts, a top surface **168** that runs the length of collecting chute **166**, and a discharge port **169**. Each duct **162** includes an inlet **163**, a chamber **164**, and an outlet **165**. Inlet **163** of each duct **162** is configured to receive undersized material from collecting pans **161** and funnel the material through chamber **164** of duct **162** to outlet **165**. Each outlet **165** communicates with a portion of top surface **168** of undersized collecting chute **166** such that material discharged from outlets **165** of ducts **162** enters collecting chute **166** and exits through discharge port **169**. An undersized material hopper may be configured to receive undersized material discharged from discharge port **169**. Though not shown, inlets **163** of ducts **162** may include radial clearances to accommodate vibratory motion from collecting pans **161** (see FIGS. **3** and **4**), which are mounted to screening deck assemblies **400**, whereas ducts **162** and collecting chute **166** are mounted to fixed outer frame **110**. The placement of the undersized collecting chutes directly beneath ducts **162** increases the efficiency of

vibratory screening machine **100** and saves space by centralizing the flow of all undersized material into a central channel.

(40) FIGS. **12**A and **12**B to FIGS. **13**A and **13**B illustrate oversized material collection assembly **170**. Oversized material collection assembly **170** includes a plurality of oversized collecting chutes **171** mounted to lower end plate **428** of each screening deck assembly **400**, and two oversized collecting troughs **176** and **176**′ in communication with oversized collecting chutes **171** (see FIGS. **3** and **4**, for example).

(41) FIGS. **12**A and **12**B illustrate an embodiment of oversized collecting chute **171**. FIGS. **13**A and **13**B illustrate an embodiment of oversized collecting trough **176**. Referring to FIGS. **12**A & **12**B, each oversized collecting chute **171** includes a first side **172** and a second side **172**′ mirroring first side **172**, both having an inlet **173** with a mounting arm **173**A, a chamber **174**, and an outlet **175**. Mounting arms **173**A of each oversized collecting chute **171** are secured to each lower endplate **428** of screening deck assemblies **400** with securing mechanisms, such as bolts, such that material that does not pass through screens **409** and/or **419** to undersized discharge assembly rolls off lower endplate **428** of screening deck assemblies **400** into inlet **173** of oversized material collecting chute **171** (see FIGS. **3** to **4**, for example). Upon or after entry into inlet **173**, oversized material is funneled through chamber **174**, and discharged from outlet **175** into oversized collecting trough **176**. While shown having a trapezoidal shape, it will be appreciated that oversized collecting chute **171** is not limited to this configuration. Oversized collecting chute **171** may have other arrangements, so long as such a chute can receive oversized material from lower endplate **428** of screening deck assemblies **400** and can transfer oversized material to one of oversized collecting troughs **176** and **176**′.

(42) Referring to FIGS. **13**A and **13**B, oversized collecting trough **176** includes a mounting end plate **177**, a back surface **178**, an outlet **180**, and a channel **181**. Mounting end plate **177** is secured to rear channel **129** of inner frame **120** with securing mechanisms, such as bolts (see FIGS. **3** and **4**, for example). Channel **181** extends from mounting end plate **177** to outlet **180** beneath each outlet **175** of oversized collecting chutes **171** such that oversized material discharged from each of oversized collecting chutes **171** falls into channel **181** of oversized collecting trough **176**. A vibratory motor **179** is mounted to back surface **178** of oversized collecting trough **176** with securing mechanisms, such as bolts, to increase the rate at which oversized material passes through channel **181** to outlet **180**, thus increasing the volume of material that vibratory screening machine **100** may process overall. Though not shown, an oversized material hopper may be configured to receive oversized materials discharged from outlet **180** of oversized collecting trough **176**.

(43) FIG. **14** is a side view similar to FIG. **7** of screening deck assembly **400** showing details of tensioning assembly **450** tensioning second screen **419** along second screening deck **420**. As indicated in FIG. **14**, material to be screened **500** flows via vibration across first screen assembly **409** toward discharge end **409**B of first screen assembly **409**. During passage, appropriately sized particles of material **500** pass through openings or pores **488**A of first screen assembly **409**. After passing over the discharge end **409**B of first screen assembly **409**B, material **500** passes into wash tray **440** and over curved side member **446** and maximum rise **449**. After passing over maximum rise **449**, the material **500** lands on an impact area **448** of second tray **419**, and then vibrates across second screen **419**, passing from input end **419**A to discharge end **419**B, with appropriately sized particles of material **500** passing through second screen **419** along the way. Screens **409**, **419** are selectively affixed to decks **410**, **420** via deck clips **455**B of the decks **410**, **420** and tensioning strips **455** of the tensioning devices **450**, in a manner described in greater detail below.

(44) As it can be understood from FIG. **14** and as is explained in further detail below, a discharge end **409**B, **419**B of screen assemblies **409**, **419** is attached to a fixed deck clip **455**B, while an opposing input end **409**A, **419**A is attached to a tensioning strip **455** of tensioning device **450**. When tensioning strip **455** is rotated, the screen assembly **409**, **419** is tensioned front-to-back across the associated deck **410**, **420**, in the same direction that material to be screened flows across

the screen deck assembly **400**. This is an improvement over earlier systems, where screen assemblies were tensioned from the sides, leaving a crown that was perpendicular to the flow of the material to be screened, creating valleys and inefficiencies in flows.
(45) FIG. **15** is a side perspective view of a screening deck assembly **400** showing additional details of first and second screen assemblies **409**, **419** tensioned over first and second screening decks **410**, **420**, respectively. In FIG. **15**, portions of screens **409**, **419** have been cutaway to show aspects of decks **410**, **420** below the screens. Material **500** is shown passing over wash tray **440** and crashing onto impact area **448** of second filter **419**.
(46) FIGS. **16**A and **16**B show views of a screen assembly **419** for use with the vibratory screening machine **100** and screening deck assembly **400** described above. While the following description of embodiments depicted in FIGS. **16**A and **16**B is made with reference to second screen assembly **419**, it is noted that this discussion applies equally to first screen assembly **409**; first screen assembly **409** can typically be identical to screen assembly **419**, but optionally may have different sizes and configurations, e.g., different sized impact area **448** (smaller or larger), different size opening configurations, a combination thereof, or the like.
(47) FIG. **16**A is a front-side perspective view of screen **419** in accordance with one or more embodiments of the disclosure. Screen **419** is configured for removably securing to deck **420** under tension in the manner described herein. Screen **419** includes feed end **419**A and opposing discharge end **419**B. Screen **419** has a widthwise dimension between ends **419**A and **419**B, and a lengthwise dimension between opposing side edges **483**. A filter area **488** is defined by a plurality of individual openings or pores **488**A extending substantially across the surface of the screen **419**. The openings **488**A are of a selected size, such as a size determined by side lengths having respective magnitudes in a range from about 20 microns and about 100 microns. In some embodiments, the openings **488**A can be rectangular shaped and can have a substantially uniform width or substantially uniform thickness in a range between about 43 microns to about 100 microns, and a substantially uniform length in a range between about 43 microns to about 2000 microns.
(48) In the embodiment of FIG. **16**A, the filter area **488** is framed by an impact zone **448** formed along feed end **419**A, a strip **486** along discharge end **419**B, and opposing side strips **484** along respective side edges **483**. Ends of the impact zone **448**, strip **486**, and side strips **484** integrally join together at abutment points, and together provide structural support to the filter area **488**, preventing tearing and the like during placement and use on the machine **100**. With reference to FIG. **14**, as material **500** flows over the curved member **446** of the wash tray **440**, the material **500** lands on impact zone **448**. Impact zone **448** protects the integrity of the individual openings **488**A and prevents or decreases the likelihood of large particles becoming lodged in the openings **488**A. As indicated in FIG. **14**, as material **500** flows from feed end **419**A to discharge end **419**B, appropriately sized particles of material **500** pass through openings **488**A. Impact zone **448** may have different sizes and configurations depending on the screening application and desired flow characteristics.
(49) As is shown in FIGS. **16**A and **16**B, a first binder strip **481**A is provided along feed end **419**A, while a second binder strip **481**B is provided along discharge end **419**B. Each binder strip **481**A, **481**B may be a generally U-shaped strip of metal that is integrated into feed ends **419**A, **419**B, substantially along the length of each respective end **419**A, **419**B. While alternative means may be used to attach binder strips **481**A, **481**B to screen **419**, the binder strips **481**A, **481**B are configured to withstand substantial forces during operation of the vibratory screening machine **100** without separating from screen **419** or otherwise allowing screen **419** to come loose from deck **420**.
(50) FIG. **16**B is a side view of a screen filter **419** for use in an exemplary embodiment of the present disclosure. When viewed from the side as in FIG. **16**B, screen **419** presents a thin profile. As seen in FIG. **16**B, the screen filter **419** includes a material input surface **485**A on an upper side, and a material output surface **485**B on an opposing lower side thereof. Individual screen openings **488**A extend from input side **485**A to output side **485**B, such that during vibratory screening,

individual particles pass through the screen area **488**. In the embodiment depicted in FIG. **16**B, first and second binder strips **481**A, **481**B depend downward from the lower side of screen **419**. Each binder strip **481**A, **481**B curves back toward a center of screen **419**, such as in an L-shape or C-shape.

(51) The screen assembly **409**, **419** is dimensioned to match the size of deck **410**, **420**. In some embodiments, screen assembly **409**, **419** preferably has a length of about 56 cm, a width of about 30 cm, and a thickness of about 0.25 cm. Impact area **448** is about 3 cm wide; narrower or wider impact areas **448** can be used, with the former decreasing protection and the latter decreasing the number of openings **488**A. Strip **486** and side strips **484** are about 1 cm wide. The screens **409**, **419** are preferably made of polyurethane. While exemplary embodiments of screens **419** are depicted in FIG. **16**A and FIG. **16**B for use with the vibratory screening machine **100** described herein, it will be appreciated that the machine **100** can be configured for use with alternative configuration of screens, screen materials, and screen characteristics (opening/pore size, connection mechanisms, and the like). Examples of screens, screen materials and screen characteristics that can be incorporated into screens **409**, **419** for use with machine **100** are found in applicant's U.S. Pat. No. 9,409,209, U.S. Patent Application Publication 2013/313,168A1, U.S. Patent Application Publication 2014/0262978A1, and U.S. Patent Application Publication 2016/0310994A1, the disclosures of which are incorporated herein by reference in their entirety.

(52) A method of attaching a screen assembly **409**, **419** to a deck **410 420** will now be described. As is seen in FIG. **14**, deck clips **455**B are fixed adjacent to respective output ends **410**B, **420**B of decks **410**, **420**. Deck clips **455**B are sized and configured for attaching output ends **409**B, **419**B of screens **409**, **419** to screening decks **410**, **420**. In an embodiment, deck clips **455**B extend substantially along a length of discharge end **410**B, **420**B, in a manner analogous to binder strips **481**A, **481**B extending along lengths of screen assembly **409**, **419**. In FIG. **14**, deck clip has an L-shaped aspect when viewed in side profile, although other engagement configurations, such as curved C-shaped aspects, can be used. As can be understood from FIG. **14**, second binder strip **481**B along discharge end **409**B, **419**B of a screen assembly **409**, **419** is engaged to deck clip **455**B, such that the L- or C-shaped aspect of binder strip **481**B interdigitates with L- or C-shaped aspect of deck clip **455**B. Tension is applied to spread screen assembly **409**, **419** across the deck **410**, **420** toward input end **410**A, **420**A, such that binder clip **481**B remains interconnected with deck clip **455**B. With screen assembly **409**, **419** spread across deck **410**, **420**, first binder strip **481**A of screen assembly **409**, **419** is then engaged to tensioning strip **455** of tensioning device **450**, such that an L- or C-shaped aspect of tensioning strip **455** interconnects with first binder strip **481**A. Tension is then applied to screen assembly **409**, **419** via tensioning device **450** to thereby selectively lock first binder strip **481**A to tensioning strip **455**, whereby filter **409**, **419** is tensioned tightly along deck **410**, **420** for use in screening particles of material **500** during operation of the machine **100**.

(53) After a period of use, screens **409**, **419** can be selectively removed from deck **410**, **420** for replacement with new screens **409**, **419**. In a method of screen removal, tensioning device **450** is used to release tension strip **455** from first strip **481**A. Screen assembly **409**, **419** is then pulled or slid toward discharge end **410**A, **420**A of deck **410**, **420** to release second binder strip **481**B from deck clip **455**B.

(54) Conditional language, such as, among others, “can,” “could,” “might,” or “may,” unless specifically stated otherwise, or otherwise understood within the context as used, is generally intended to convey that certain implementations could include, while other implementations do not include, certain features, elements, and/or operations. Thus, such conditional language generally is not intended to imply that features, elements, and/or operations are in any way required for one or more implementations or that one or more implementations necessarily include logic for deciding, with or without user input or prompting, whether these features, elements, and/or operations are included or are to be performed in any particular implementation.

(55) This specification and annexed drawings disclose vibratory screening machines that include

stacked screening deck assemblies. It is, of course, not possible to describe every conceivable combination of elements for purposes of describing the various aspects of the disclosure. Thus, while embodiments of this disclosure are described with reference to various implementations and exploitations, it is noted that such embodiments are illustrative and that the scope of the disclosure is not limited to them. Those of ordinary skill in the art can recognize that many further combinations and permutations of the disclosed features are possible. As such, various modifications can be made to the disclosure without departing from the scope or spirit thereof. In addition or in the alternative, other embodiments of the disclosure can be apparent from consideration of the specification and annexed drawings, and practice of the disclosure as presented herein. It is intended that the examples put forward in the specification and annexed drawings be considered, in all respects, as illustrative and not restrictive. Although specific terms are employed herein, they are used in a generic and descriptive sense only and not for purposes of limitation.

Claims

1. A vibratory screening machine, comprising: an outer frame; an inner frame connected to the outer frame; a vibratory motor assembly attached to the inner frame such that the vibratory motor assembly vibrates the inner frame; a plurality of screen deck assemblies attached to the inner frame and configured in a stacked arrangement, each one of the plurality of screen deck assemblies configured to receive replaceable screen assemblies, the replaceable screen assemblies secured to the screen deck assemblies by tensioning the replaceable screen assemblies in a direction that a material to be screened flows across the replaceable screen assemblies; an undersized material discharge assembly configured to receive materials that pass through the replaceable screen assemblies; and an oversized material discharge assembly configured to receive materials that pass over a top surface of the replaceable screen assemblies.
2. The vibratory screening machine of claim 1, further comprising a screen tensioning system that includes a plurality of tensioning rods that extend substantially orthogonal to the direction of flow of the material being screened, wherein each tensioning rod is configured to mate with a portion of a replaceable screen assembly and to tension the replaceable screen assembly when rotated.
3. The vibratory screening machine of claim 2, wherein the screen tensioning system includes a plurality of ratcheting assemblies, wherein each ratcheting assembly is configured to rotate a corresponding one of the plurality of tensioning rods such that the tensioning rod moves between a first open screen assembly receiving position and a second closed and secured screen assembly tensioned position.
4. The vibratory screening machine of claim 1, wherein each screen deck assembly includes: a first replaceable screen assembly; a second replaceable screen assembly; a first tensioning rod configured to mate with a portion of the first replaceable screen assembly and to tension the first replaceable screen assembly when rotated; and a second tensioning rod configured to mate with a portion of the second replaceable screen assembly and to tension the second replaceable screen assembly when rotated; wherein the first and second tensioning rods can be independently rotated to independently tension of the first and second replaceable screen assemblies.
5. The vibratory screening machine of claim 1, wherein each screen deck assembly includes a first replaceable screen assembly and a second replaceable screen assembly, and wherein the first and second replaceable screen assemblies of each screen deck assembly can be independently tensioned in a direction that a material to be screened flows across the replaceable screen assemblies.
6. The vibratory screening machine of claim 1, wherein the undersized material discharge assembly includes an undersized chute in communication with each one of the plurality of screen deck assemblies, and wherein the oversized material discharge assembly includes an oversized chute assembly in communication with each one of the plurality of screen deck assemblies.
7. The vibratory screening machine of claim 6, wherein the oversized chute assembly includes a

first oversized chute assembly in communication with each of the plurality of screen deck assemblies and a second oversized chute assembly in communication with each of the plurality of screen deck assemblies.

8. The vibratory screening machine of claim 7, wherein the undersized chute, the first oversized chute assembly, and the second oversized chute assembly are located beneath the plurality of screen deck assemblies, and wherein the undersized chute is located between the first oversized chute assembly and the second oversized chute assembly.

9. The vibratory screening machine of claim 7, wherein the oversized chute assembly includes a bifurcated trough that is configured to receive materials that do not pass through the replaceable screen assemblies and that are conveyed over a discharge end of the plurality of screen deck assemblies, a first section of the bifurcated trough feeding the first oversized chute assembly and a second section of the bifurcated trough feeding the second oversized chute assembly.

10. A screen deck assembly, comprising: a first screen deck configured to receive a first screen assembly; a second screen deck configured to receive a second screen assembly and located downstream from the first screen deck; a screen tensioning system that includes first and second tensioning rods that extend substantially orthogonal to a direction of flow of a material to be screened, wherein the first tensioning rod is configured to mate with a portion of a first screen assembly located on the first screen deck and to tension the first screen assembly in a direction that material to be screened flows across the first screen assembly when rotated to secure the first screen assembly to the first screen deck, and wherein the second tensioning rod is configured to mate with a portion of a second screen assembly located on the second screen deck and to tension the second screen assembly in a direction that material to be screened flows across the screen assembly when rotated to secure the second screen assembly to the second screen deck.

11. The screen deck assembly of claim 10, and wherein the first and second tensioning rods are independently rotatable such that first and second screen assemblies on the first and second screen decks can be independently tensioned.

12. The screen deck assembly according to claim 11, wherein the screen tensioning system further comprises: a first ratcheting assembly configured to rotate the first tensioning rod such that the first tensioning rod moves between a first open screen assembly receiving position and a second closed and secured screen assembly tensioned position; and a second ratcheting assembly configured to rotate the second tensioning rod such that the second tensioning rod moves between a first open screen assembly receiving position and a second closed and secured screen assembly tensioned position.

13. A method of screening a material, comprising: feeding the material on a vibratory screening machine having a plurality of screen deck assemblies that are configured in a stacked arrangement, each one of the plurality of screen deck assemblies configured to receive replaceable screen assemblies, the replaceable screen assemblies secured to the plurality of screen deck assemblies by tensioning the replaceable screen assemblies in a direction the material flows across the replaceable screen assemblies; and screening the materials such that an undersized material that passes through the replaceable screen assemblies flows into an undersized material discharge assembly and an oversized material flows over an end of the plurality of screen deck assemblies into an oversized material discharge assembly.

14. The method of screening a material according to claim 13, wherein the vibratory screening machine further comprises a screen tensioning system that includes a plurality of tensioning rods that extend substantially orthogonal to the direction of flow of the material being screened, wherein each tensioning rod is configured to mate with a portion of a replaceable screen assembly and to tension the replaceable screen assembly when rotated.

15. The method of screening a material according to claim 14, wherein each one of the plurality of screen deck assemblies includes: a first screen deck configured to receive a first replaceable screen assembly, the first screen deck including a first tensioning rod configured to mate with a portion of

a first replaceable screen assembly located on the first screen deck and to tension the first replaceable screen assembly when rotated to secure the first replaceable screen assembly to the first screen deck; and a second screen deck configured to receive a second replaceable screen assembly, the second screen deck including a second tensioning rod configured to mate with a portion of a second replaceable screen assembly located on the second screen deck and to tension the second replaceable screen assembly when rotated to secure the second replaceable screen assembly to the second screen deck.

16. The method of screening a material according to claim 15, wherein the first and second tensioning rods are independently rotatable.

17. The method of screening a material according to claim 13, wherein the undersized material discharge assembly includes an undersized chute in communication with each one of the plurality of screen deck assemblies and the oversized material discharge assembly includes an oversized chute assembly in communication with each one of the plurality of screen deck assemblies.

18. The method of screening materials according to claim 17, wherein the oversized chute assembly includes a first oversized chute assembly and a second oversized chute assembly.

19. The method of screening a material according to claim 18, wherein the undersized chute, the first oversized chute assembly, and the second oversized chute assembly are located beneath the plurality of screen deck assemblies, and wherein the undersized chute is located between the first oversized chute assembly and the second oversized chute assembly.

20. A vibratory screening machine for screening particles of a material to be screened, comprising: an outer frame; an inner frame connected to the outer frame; a vibratory motor assembly secured to the inner frame such that the vibratory motor assembly vibrates the inner frame; a plurality of screen deck assemblies attached to the inner frame and configured in a generally stacked arrangement, each of the plurality of screen deck assemblies having a front-to-back dimension extending from a material input end to a material output end, each of the screen deck assemblies being configured such that a replaceable screen can be secured to the screen deck assembly by tensioning the replaceable screen along the front-to-back dimension; an undersized material discharge assembly configured to receive particles of said material that pass through the replaceable screens; and an oversized material discharge assembly configured to receive particles of said material that pass over a top surface of the replaceable screens.

21. The vibratory screening machine of claim 20, further comprising a screen tensioning system that includes a plurality of tensioning rods that extend substantially orthogonal to the front-to-back dimension, wherein each tensioning rod is configured to mate with a portion of a replaceable screen and to tension the replaceable screen when rotated, and wherein the tensioning rods are independently rotatable.

22. The vibratory screening machine of claim 21, wherein the screen tensioning system includes a plurality of ratcheting assemblies, wherein each ratcheting assembly is configured to rotate a corresponding one of the plurality of tensioning rods such that the tensioning rod moves between a first screen assembly receiving position and a second closed and secured screen assembly tensioned position.

23. The vibratory screening machine of claim 20, wherein each screen deck assembly includes: a first screen deck configured to receive a first replaceable screen; a second screen deck configured to receive a second replaceable screen; a first tensioning rod configured to mate with a portion of a first replaceable screen located on the first screen deck and to tension the first replaceable screen when rotated to secure the first replaceable screen to the first screen deck; and a second tensioning rod configured to mate with a portion of a second replaceable screen located on the second screen deck and to tension the second replaceable screen when rotated to secure the second replaceable screen to the second screen deck; wherein the first and second tensioning rods can be independently rotated to independently tension of the first and second replaceable screens.

24. The vibratory screening machine of claim 20, wherein the undersized material discharge

assembly includes an undersized chute in communication with each of the plurality of screen deck assemblies, and wherein the oversized material discharge assembly includes an oversized chute assembly in communication with each of the plurality of screen deck assemblies.

25. The vibratory screening machine of claim 24, wherein the oversized chute assembly includes a first oversized chute assembly in communication with each of the plurality of screen deck assemblies and a second oversized chute assembly in communication with each of the plurality of screen deck assemblies.

26. The vibratory screening machine of claim 25, wherein the undersized chute, the first oversized chute assembly and the second oversized chute assembly are located beneath the plurality of screen deck assemblies, and wherein the undersized chute is located between the first oversized chute assembly and the second oversized chute assembly.